

Product Carbon Footprint Study Report

Product Name: battery production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: battery production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg battery production, Li-ion, LFP, rechargeable

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg battery production, Li-ion, LFP, rechargeable.

2. System Boundary

battery production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of battery production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **12.829 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

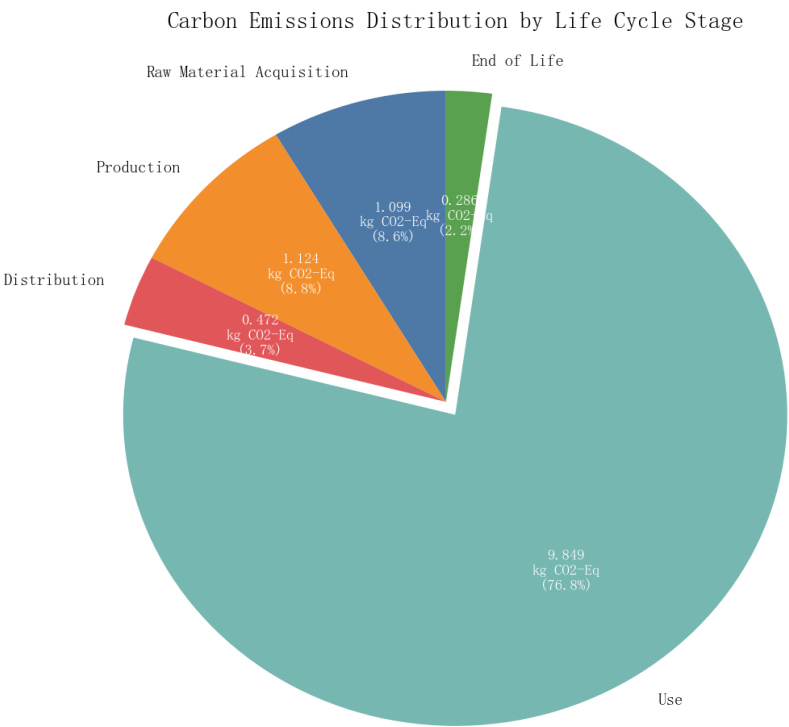


Table 1: Carbon Emissions by Life Cycle Stage

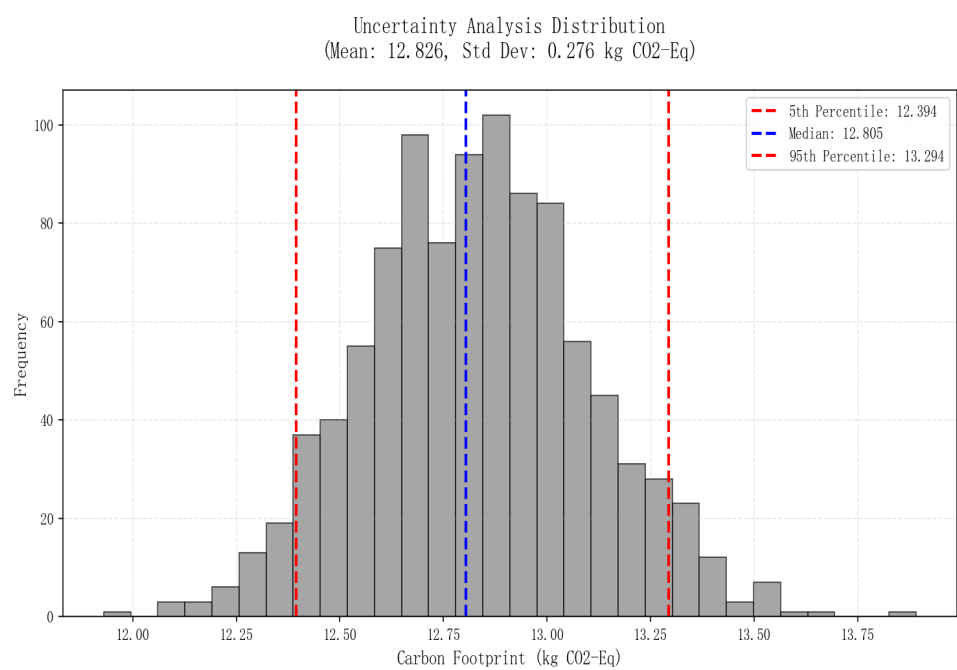
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	1.099	8.57
Production	1.124	8.76
Distribution	0.472	3.68

Use	9.849	76.77
End of Life	0.286	2.23
Total	12.829	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	12.826
Standard Deviation	0.276
Coefficient of Variation (CV)	2.16%
Median	12.805
5th Percentile	12.394
95th Percentile	13.294

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the battery production system is 12.829 kg CO₂-Eq, representing a moderate emission intensity for this product category when assessed over its full life cycle. This value reflects the aggregated greenhouse gas emissions from raw material extraction through end-of-life treatment, providing a benchmark for environmental performance evaluation.

Lifecycle stage contributions reveal a highly skewed distribution, with the use phase dominating at 9.849 kg CO₂-Eq (76.8%). Raw material and production stages contribute 8.6% and 8.8% respectively, while distribution and end-of-life phases remain marginal at under 4% each. This pattern indicates that operational energy consumption during the battery's service life constitutes the primary emission driver.

Major emission sources are concentrated in the use stage, followed by upstream manufacturing processes. The dominance of the use phase suggests that electricity or fuel inputs during battery operation are the critical hotspots requiring targeted mitigation strategies.

2. Statement of Assumptions and Limitations

Key assumptions in this study include standardized operational profiles for battery usage, average electricity grid mixes, and typical transportation distances for distribution. These assumptions enable consistent modeling but may not fully capture regional variations in user behavior or energy sources.

Limitations arise from reliance on secondary data sources such as generic life cycle inventory databases, which may introduce uncertainties due to temporal mismatches or aggregated industrial averages. Data quality for foreground processes is moderate, while background processes depend on older datasets that could overestimate or underestimate emissions.

System boundary restrictions encompass a cradle-to-grave scope excluding potential recycling credits beyond end-of-life treatment and omitting indirect land-use changes or infrastructure impacts. The analysis is confined to carbon footprint metrics without extending to other impact categories.

3. Suggestions for Improvement

- Optimize battery energy density and efficiency to reduce electricity consumption in the use phase, potentially lowering emissions by 20-30% through advanced cell chemistries.
- Transition raw material sourcing to low-carbon suppliers and increase recycled content in electrodes to cut upstream emissions by 15%.
- Implement renewable energy in production facilities to address the 8.8% contribution from manufacturing processes.
- Enhance end-of-life collection and recycling infrastructure to recover critical materials and minimize residual emissions.

- Improve primary data collection via supplier-specific inventories to enhance overall data quality and reduce uncertainty in future assessments.

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